

Review Assist Environmental Constraints Extraction Package

Prepared 2026-05-21

Implementation-oriented design package from uploaded Review Assist framing notes and example Environmental Constraints Report

Source posture

This package extracts reusable design intelligence from the uploaded Review Assist framing notes and example Environmental Constraints Report. The example report is treated as a working draft and target-shape reference, not a finished template or a source of reusable project facts. Project-specific, trail/corridor-specific, PEL-specific, incomplete, and restricted/manual material are explicitly separated from generalizable policy.

Executive implementation stance

Review Assist should become a configurable environmental constraints report engine where every section knows its source requirements, evidence grain, spatial extent, visual extent, figure/table support, caveat bundle, prohibited claims, comparison-unit behavior, drafting mode, and review gate. The first implementation move should be a canonical machine-readable policy object that blocks sections from rendering unless policy triggers and source conditions are satisfied.

Package contents

- Main memo in DOCX, PDF, and Markdown
- Canonical policy starter JSON
- Concise policy explanation
- GPT style-context file
- Codex prompt starters

1. Report Structure Inventory

Classification intentionally separates generalizable report shape from conditional modules, manual/restricted material, PEL-specific text, trail/corridor behavior, and draft-only artifacts.

Table 1. Report structure inventory

Section	Purpose	Classification	Output shape	Backing type	Default?	Trigger / omit logic
Cover / title	Package identity	Generalizable shape; project-specific content	Cover metadata	Project metadata	Yes	Always; generated from project metadata
Executive Summary	High-level constraints summary	Generalizable shape; generated after review	Compact narrative	Reviewed evidence + reviewer approval	Yes	Render only after sections/tables reviewed
TOC / lists	Document navigation	Generalizable	Auto-generated/static lists	Presentation only	Yes	Generate when corresponding figures/tables/attachments exist
1 Introduction	Introduce review purpose and project context	Generalizable, but narrative is custom	Paragraph	Reviewer/project metadata	Yes	Always, but keep neutral and compact
1.1 Relationship with PEL Study	Tie report to a PEL process	PEL-specific / conditional / manual	Manual narrative	Reviewer-written	No	Only if parent_study.type is PEL or reviewer adds it
1.2 Study Area / Project Area	Describe project geography and comparison units	Generalizable concept; trail-specific sample execution	Paragraph + optional map/list	Project geometry + metadata	Yes	Use neutral project area / comparison units language
2 Methodology	Explain source, mapping, and limits	Generalizable	Short narrative + source refs	Source manifest + app config	Yes	Always; reflect actual app workflow
2.1 Data Collection and Sources	Name data types and provenance	Generalizable	Paragraph + source list/ref	Source manifest	Yes	Always; generated from sources actually used
2.2 Mapping and Analysis Procedures	Explain overlays, basemaps, panels	Generalizable; tool names are app-specific	Paragraph/bullets	Figure policy + app config	Yes	Do not copy ArcGIS version unless true
2.3 Limitations and Data Gaps	Desktop-review caveats	Generalizable	Caveat paragraph/list	Caveat library + source status	Yes	Always; insert source-specific caveats
3 Environmental Constraints Inventory	Parent inventory section	Generalizable	Intro + attachment ref	Section matrix	Yes	Include if any modules are enabled
3.1 Natural and Ecological Resources	Parent natural resources category	Generalizable family	Short intro	Source-backed	Conditional	Include when natural modules are enabled
3.1.1 Wetlands and Waterbodies	NWI wetlands, streams, ponds, crossings	Generalizable module; per-alt narrative conditional	Paragraph + table + figure	Source-backed	Conditional/default if sources exist	Trigger on NWI/hydrography sources
3.1.1.x Alternative subsections	Route-specific wetlands narratives	Trail/corridor-specific + incomplete	Narrative children	Source-backed + design assumptions	No	Only if reviewer enables narrative comparison
3.1.2 Floodplains and Floodways	FEMA flood zones/floodways	Generalizable module	Paragraph + table + figure	Source-backed	Conditional/default if source exists	Trigger on FEMA/NFHL/DFIRM source
3.1.3 Water Quality	HUC, 303(d), TMDL, downstream context	Conditional/generalizable	Paragraph + watershed map	Source-backed	Conditional	Trigger on impaired waters/TMDL/watershed source or relevance
3.1.4 Protected Species and Critical Habitat	Agency/source species summary	Conditional/manual	Paragraph + letter/source ref	Agency/reviewer supplied	No default	Include with IPaC/state source or agency letters
3.2 Cultural and Historic Resources	Cultural compliance context	Conditional; sensitive data	Short intro + restricted refs	Restricted/manual/public context separated	Conditional	Only with cultural module or manual source
3.2.1 Archaeological Sites	APE and recorded sites	Restricted/manual	Manual paragraph/attachment	Authorized records	No default public drafting	Only with authorized records and reviewer-defined APE
3.2.2 Historic Structures and Districts	Architectural APE and older properties	Conditional/manual	Paragraph/map/table if authorized	Authorized records + optional property age	Conditional	Include with architecture/cultural source
3.3 Community Resources	Access-sensitive community facilities	Conditional	Intro + lists/maps	Source-backed + reviewer context	Conditional	Trigger on community-facility sources or project relevance
3.3.1 Fire/EMS Stations	Emergency access sensitivity	Conditional	Paragraph/list/map	Source-backed + review	Conditional	Trigger on nearby facilities
3.3.2 Government Buildings	Public-service access	Conditional	Paragraph/list/map	Source-backed	Conditional	Trigger on nearby facilities
3.3.3 Education Facilities	School/childcare access	Conditional	Paragraph/list/map	Source-backed	Conditional	Trigger on nearby schools/childcare
3.3.4 Health Care Facilities	Medical access/noise sensitivity	Conditional	Paragraph/list/map	Source-backed + review	Conditional	Trigger on nearby healthcare facilities
3.3.5 Places of Worship	Community-gathering access	Conditional	Paragraph/list/map	Source-backed, often coarse	Conditional	Trigger on nearby facilities; avoid exhaustive claims
3.3.6 Parks and Recreation	Recreation access/context	Conditional	Paragraph/list/map	Source-backed	Conditional	Trigger on nearby/direct

Section	Purpose	Classification	Output shape	Backing type	Default?	Trigger / omit logic
Areas						parks/recreation resources
3.4 Utility and Infrastructure Considerations	Utility/infrastructure context	Conditional	Intro + lists/maps	Source-backed/manual	Conditional	Include if utility/infrastructure sources or reviewer flags exist
3.4.1 Public Water Supply	Wells/water facilities	Conditional	Paragraph + map	Source-backed	Conditional	Trigger on nearby wells/facilities
3.4.2 Utility Infrastructure	Pipelines/utilities	Conditional; sensitive	Paragraph/list/map	Source-backed + provider coordination	Conditional	Trigger on crossings/nearby assets; sensitivity rules
3.4.3 Energy Infrastructure	Transmission lines/substations	Conditional	Paragraph/list/map	Source-backed + provider coordination	Conditional	Trigger on nearby/intersecting assets
3.4.4 Airports	Airport proximity	Conditional	Short paragraph/map if relevant	Source-backed	No default	Trigger if airport within threshold or aviation relevance
3.5 Contamination Risks	Regulatory/hazmat context	Generalizable/conditional	Intro	Source-backed/manual	Conditional/default if hazmat enabled	Include if regulated-facility sources configured
3.5.1 Hazardous Materials Sites	Facility database/risk summary	Generalizable module; risk ranking review-gated	Paragraph + table + map + attachment ref	Source-backed + specialist/reviewer	Conditional/default if sources exist	Risk labels require policy/review
3.5.2 Oil Wells	Oil/gas well screening	Conditional by geography/source	Short paragraph/table/map if findings	Source-backed	Conditional	Include where oil/gas source configured
3.6 Socioeconomic and Business Considerations	Demographics and economic context	Conditional	Intro + tables/prose	Source-backed + manual	Conditional	Trigger on planning/access/community context
3.6.1 Demographic Characteristics	Census tract/community context	Conditional/generalizable	Tables + short interpretation	Source-backed	Conditional	Trigger on Census/ACS source
3.6.2 Local Businesses and Economic Nodes	Commercial access context	Project-specific/manual	Manual narrative/list/map optional	Reviewer/local context	No default	Only if reviewer supplies or business source enabled
4 Conclusion and Next Steps	Reviewed summary and action path	Generalizable shape; reviewer judgment	Narrative	Reviewed evidence + decisions	Yes but gated	Generate after review; no new findings
5 References	Bibliography of used sources	Generalizable	Reference list	Source manifest	Yes	Only cite sources actually used
Attachments	Maps, hazmat report, agency letters	Generalizable package model; contents conditional	Attachment list/package	Figure matrix + manual sources	Conditional	Include only generated/supplied attachments

2. Section Policy Table

This matrix should become the seed for the canonical Review Assist policy file. Every report section should carry these fields before deterministic or GPT drafting is allowed.

Table 2. Section policy matrix

section_id	Family	Extent type	Visual extent	Comparison-unit policy	Evidence pattern	Drafting mode	Output shape
exec_summary	summary	presentation-only	presentation-only	none	reviewed section summaries	GPT only after reviewer confirmation	Compact narrative
intro.project_overview	intro	manual/project metadata	presentation-only	context_summary_list	project metadata	GPT after reviewer confirmation	Paragraph
intro.parent_study_relation ship	planning context	manual/reviewer-supplied	presentation-only	manual_only	reviewer-supplied material	manual only	Paragraph
intro.project_area_compari son_units	project geography	direct project/comparison-unit extent	medium/context	context_summary_list	project geometry + comparison-unit metadata	GPT after reviewer confirmation	Paragraph + optional map
method.sources	methodology	presentation-only	presentation-only	none	source status/table summary	deterministic only	Paragraph + source appendix ref
method.mapping_analysis	methodology	presentation-only	presentation-only	none	figure/map support	deterministic only	Paragraph/bullets
method.limitations	limitations	presentation-only	presentation-only	none	caveat library + source status	deterministic only	Caveat paragraph/list
inventory.overview	inventory	presentation-only	attachment/panel	context_summary_list	table summary + figure support	deterministic + reviewed polish	Section intro
natural.wetlands_waterbod ies	natural	direct extent; optional screening buffer	small/direct	table_only default	direct intersections; crossing counts; table	GPT with source-backed evidence	Paragraph + table + figure

section_id	Family	Extent type	Visual extent	Comparison-unit policy	Evidence pattern	Drafting mode	Output shape
natural.floodplains_floodways	natural	direct project/comparison-unit extent	small/direct	table_only	summary; map support		
natural.water_quality	natural	watershed/subwatershed + downstream context	large/watershed	context_summary_list	direct intersections; acreage by zone; figure support	GPT with source-backed evidence	Paragraph + table + figure
natural.protected_species	natural	county/regional; screening buffer; manual	county/regional or attachment	manual_only	watershed membership; downstream context; impaired-water list	GPT with source-backed evidence	Paragraph + watershed map
cultural.archaeology	cultural	APE/direct disturbance; manual	attachment/panel	manual_only	agency/reviewer material; optional screening	GPT only after reviewer confirmation	Paragraph + letter ref
cultural.architectural	cultural	APE + visual/noise/vibration context	medium/context or attachment	manual_only	authorized records; reviewer material	manual only	Paragraph/attachment
community.fire_ems	community	nearby/community context	medium/context	context_summary_list	authorized records; optional property age	manual or GPT after confirmation	Paragraph/map/table if authorized
community.government	community	nearby/community context	medium/context	context_summary_list	nearest-within-buffer; count/list	GPT with evidence + review	Paragraph/list/map
community.education_child_care	community	nearby/community context	medium/context	context_summary_list	nearest-within-buffer; count/list	GPT with evidence + review	Paragraph/list/map
community.healthcare	community	nearby/community context	medium/context	context_summary_list	nearest-within-buffer; count/list	GPT with evidence + review	Paragraph/list/map
community.places_of_worship	community	nearby/community context	medium/context	context_summary_list	nearest-within-buffer; count/list	GPT with evidence + review	Paragraph/list/map
community.parks_recreation	community	direct/adjacent + nearby context	medium/context	context_summary_list	nearest-within-buffer; count/list	GPT with evidence + review	Paragraph/list/map
infrastructure.public_water_supply	infrastructure	nearby context; optional buffer	medium/context	context_summary_list	nearest-within-buffer; figure support	GPT with evidence + review	Paragraph/map
infrastructure.utilities	infrastructure	direct intersections + nearby context	small/direct or medium/context	table_only or context_summary_list	direct intersections; list context	GPT only after reviewer confirmation	Paragraph/list/map
infrastructure.energy	infrastructure	direct intersections + nearby context	small/direct or medium/context	table_only or context_summary_list	direct intersections; list context	GPT only after reviewer confirmation	Paragraph/list/map
infrastructure.airports	infrastructure	county/regional; nearest-within-buffer	county/regional	none	nearest-within-buffer	deterministic or GPT with evidence	Short paragraph
contamination.hazmat_sites	contamination	screening buffer + direct extent	medium/context	table_only + facility list	nearest-within-buffer; risk table; attachment	GPT with evidence; risk labels review-gated	Paragraph + table + map
contamination.oil_gas_wells	contamination	direct extent + screening buffer	small/direct	table_only if findings	direct intersections/nearest	deterministic or GPT with evidence	Short paragraph/map
socio.demographics	socioeconomic	census/community + county/regional comparison	county/regional	none	table summary; census membership	GPT with source-backed evidence	Paragraph + tables + map
socio.business_nodes	socioeconomic	nearby/community; manual	medium/context	manual_only	reviewer-supplied; optional POI context	manual or GPT after confirmation	Paragraph/list
conclusion.next_steps	conclusion	presentation-only	presentation-only	none	reviewed findings only	GPT only after reviewer confirmation	Narrative
references	references	presentation-only	presentation-only	none	source manifest	deterministic only	Reference list

Required caveats and prohibited claims

Table 3. Caveats, prohibited claims, and allowed refs

Section	Required caveats	Prohibited claims	Allowed refs
exec_summary	Screening-level; human-reviewed; based on available sources	New facts not in reviewed sections; final impact/no-impact conclusions	Reviewed sections, tables, figures, source status
natural.wetlands_waterbodies	NWI non-jurisdictional; field delineation required	Jurisdictional wetland; no wetlands; no impacts without review	NWI, NHD, wetlands table, wetlands figure
natural.floodplains_floodways	FEMA mapped hazard; local/H&H coordination design-dependent	Permit determination; flood risk guarantee	FEMA/NFHL/DFIRM table and figure

Section	Required caveats	Prohibited claims	Allowed refs
natural.water_quality	Direct vs downstream distinction; BMP/permit planning only	Direct impairment if only downstream; pollutant causation	NHD, HUC, 303(d)/TMDL source, watershed figure
natural.protected_species	Agency/source date; habitat/effect confirmation needed	No effect; no species present; no suitable habitat without confirmation	Agency letters, IPaC/state source
cultural.*	Authorized vs public/coarse source separation; reviewer-defined APE	Public context as MDAH/SHPO; eligibility/effect without authorized records	Restricted records, reviewer letters, public context labeled
community.*	Facility lists may not be exhaustive; planning context only	All facilities; safe access ensured; unsupported operational impacts	Community source table and maps
infrastructure.*	Utility data may be incomplete/sensitive; owner coordination needed	Conflict resolved; exact sensitive details if restricted	Utility source, owner data, maps
contamination.hazmat_sites	Database screening; risk tiers reviewed; not Phase I unless report supplied	Cleanup status overclaim; no contamination; site safe	EPA/state datasets, hazmat table, Appendix B
socio.demographics	Census geographies extend beyond project; context only	Disproportionate impact/EJ determination without analysis	ACS/Decennial, census map, demographic tables
conclusion.next_steps	No new evidence; reviewer judgment	Binding commitments; final regulatory findings	Reviewed sections and review decisions

3. Source Needs by Section

The source manifest should decide whether a section can render, what caveats apply, and whether data is public, coarse, restricted, manual, optional, or deferred.

Table 4. Source needs by section

Section	Source	Category	Likely public/source candidate	Need type	Supports	Output support	Current app coverage	Gap?
Project area	Project geometry/comparison units	Project metadata	Reviewer-uploaded geometry	Required/reviewer-supplied	Direct analysis	Prose, maps, comparison units	Covered conceptually	Needs comparison-unit policy
Basemaps	Aerial imagery/topographic data	Basemap	NAIP/topo/terrain services	Required/optional	Presentation/support	Figures	Covered via NAIP sidecars	Needs source-status display
Wetlands/waterbodies	USFWS NWI	Wetlands	USFWS NWI	Required for module	Direct + nearby	Table, figure, prose, caveat	Covered	Needs caveat + table policy
Wetlands/waterbodies	NHD streams/waterbodies	Hydrography	USGS NHD/NHDPlus/local hydro	Required for crossings	Direct intersections	Table/prose	Covered	Needs crossing-count policy
Floodplains	FEMA DFIRM/NFHL	Flood hazard	FEMA NFHL/DFIRM	Required for module	Direct intersections	Table, figure, prose	Covered	Needs floodway/zone acreage policy
Water quality	HUC-12 subwatersheds	Hydrology context	USGS WBD/HUC	Required for water-quality context	Watershed/subwatershed	Figure/prose	Partial via hydrography	Gap if WBD not materialized
Water quality	303(d) impaired waters	Water quality	State impaired-waters/TMDL lists	Required if module	Watershed/downstream	Prose, figure, caveat	Not listed	Gap
Water quality	TMDLs/stressors	Water quality	State TMDL database	Optional/conditional	Downstream context	Prose/caveat	Not listed	Gap
Protected species	USFWS/IPaC/critical habitat	Species/habitat	USFWS IPaC/critical habitat	Manual/optional source	Regional/range + review	Prose, letter attachment	Not listed	Gap/manual
Protected species	State species/natural heritage	Species/habitat	State agency/natural heritage	Restricted/manual	Buffer/regional	Prose, letter attachment	Not listed	Gap/manual
Cultural	MDAH/SHPO inventory	Cultural/restricted	Authorized SHPO/MDAH records	Restricted/reviewer-supplied	APE/direct/nearby	Prose/restricted figure/attachment	Future placeholder	Needs strict policy
Cultural	Public/coarse cultural context	Cultural/public context	MARIS/public layers/OpenContext/DINA A-style	Optional/public-coarse	Screening context only	Caveat/prose	Covered conceptually	Needs guardrails
Cultural	APE geometry	Project/reviewer	Reviewer-defined APE	Required for cultural module	Manual/direct	Prose/map	Not explicit	Gap/manual
Community resources	Fire/EMS, government, schools, childcare, healthcare, worship, parks	Community facilities	HIFLD, local GIS, NCES, state licensing, OSM, PAD-US	Optional/conditional	Nearby/community	List/map/prose	Not listed	Gap
Public water supply	Water treatment plant/public wells	Infrastructure/water	State health/environmental data; SDWIS context	Optional/conditional	Nearby context	Figure/prose	Not listed	Gap

Section	Source	Category	Likely public/source candidate	Need type	Supports	Output support	Current app coverage	Gap?
Utilities/energy	Pipelines/transmission/substations	Utility/energy	NPMS/HIFLD/state/local/owner supplied	Conditional/sensitive/manual	Direct + nearby	List/map/prose	Not listed	Gap/manual
Airports	Nearby airports	Transportation infrastructure	FAA airport data	Optional/conditional	Regional/nearest	Prose/map	Not listed	Gap
Hazmat	EPA/state regulatory databases	Regulated/hazardous	EPA FRS, Brownfields, NPDES, Superfund, TRI, UST, solid waste	Required for module	Screening buffer/nearby	Table, figure, prose	Covered well as class	Needs risk-ranking policy
Hazmat	Hazardous Materials Assessment Report	Technical report	Reviewer-supplied report	Manual/conditional	Attachment support	Attachment/body summary	Evidence package supports	Needs attachment workflow
Oil/gas	State oil/gas well database	Oil/gas	State oil/gas board wells	Conditional/state-specific	Direct/screening buffer	Prose/map/table	Covered as source class	Needs state source config
Demographics	Census tracts and ACS/Decennial data	Socioeconomic	Census TIGER/ACS/Decennial	Conditional	Community/regional	Map/table/prose	Not listed	Gap
Business nodes	Commercial districts/local businesses	Socioeconomic/economic	Reviewer knowledge, parcels, POI, business registry	Manual/optional	Nearby/community	Prose/list/map	Parcels optional only	Gap/manual
Agency letters	Agency consultation letters	Manual agency material	Reviewer upload	Manual/reviewer-supplied	Attachment support	Attachment + summary	Review queue supports	Needs source type + citation policy

4. Figure Needs

Figures should be matrix-backed. A figure can support evidence, but the visual extent must never overwrite the analysis extent.

Table 5. Figure needs

Figure purpose/title	Section	Likely source layers	Extent	Body/attachment	Matrix-backed?	Requirement	Evidence relation	Current support
Project location / study area / comparison units	Intro	Project geometry, basemap, jurisdictions	medium/context	Main body	Yes	Recommended default	Presentation + direct geometry	Likely supported; needs policy
Overall Environmental Constraints Inventory Map	Inventory / Attachment A	All active constraints layers	large/project-wide	Attachment/support	Yes	Conditional	Presentation/support	Figure generation supports class
Panel Index Map	Attachment A	Project extent, panel footprints	attachment/panel	Attachment	Yes	Conditional	Presentation/support	Needs panel policy
Detailed Panel Maps	Attachment A	All active constraints layers + basemap	attachment/panel	Attachment	Yes	Conditional, especially long corridors	Presentation/support	App supports figures; needs paneling
Wetlands and Waterbodies	3.1.1	NWI, NHD/waterbodies, project geometry, basemap	small/direct or medium/context	Main body	Yes	Conditional/default if source enabled	Direct evidence + nearby context	Supported
FEMA Flood Zones	3.1.2	FEMA/NFHL/DFIRM, project geometry	small/direct	Main body	Yes	Conditional/default if source enabled	Direct evidence	Supported
Streams and 303(d) Impaired Waters within Subwatersheds	3.1.3	NHD, WBD/HUC-12, 303(d), TMDL, project geometry	large/watershed	Main or attachment	Yes	Conditional	Watershed/downstream context	Hydro supported; 303(d)/WBD gap
Protected species / critical habitat	3.1.4	Critical habitat/range/source letters/project geometry	county/regional or attachment	Usually attachment/manual	Yes if source exists	Conditional/manual	Screening context	Gap
Cultural resources	3.2	Restricted cultural records, APE, project geometry	attachment/restricted	Attachment/restricted	Yes with access controls	Manual/restricted	Authorized records	Placeholder only; needs policy
Community resource maps	3.3.x	Fire/EMS, government, schools, childcare, healthcare, worship, parks	medium/context	Main or attachment	Yes	Conditional	Nearby/community context	Gap
Public Water Supply Wells	3.4.1	Public wells, water facilities, project geometry	medium/context	Main or attachment	Yes	Conditional	Nearby infrastructure context	Gap
Utility/Energy Infrastructure	3.4.2-3.4.3	Pipelines, transmission lines, substations	small/direct or medium/context	Main or attachment with sensitivity controls	Yes	Conditional/manual	Direct crossings + nearby context	Gap
Airports	3.4.4	FAA airports, project geometry	county/regional	Usually no figure unless triggered	Yes	Optional	Nearest/regional context	Gap
Hazardous/Regulated Sites	3.5.1	EPA/state hazardous layers, project buffer	medium/context	Main + support appendix	Yes	Conditional/default if hazmat enabled	Nearby/screening-buffer evidence	Source class supported

Figure purpose/title	Section	Likely source layers	Extent	Body/attachment	Matrix-backed?	Requirement	Evidence relation	Current support
Oil/Gas Wells	3.5.2	State oil/gas wells, project geometry/buffer	small/direct	Optional main/attachment	Yes	Conditional	Direct/screening evidence	Source class supported
Census Tracts	3.6.1	Census tracts, project geometry	county/regional	Main body	Yes	Conditional	Community/regional context	Gap
Business/economic nodes	3.6.2	Parcels, POI/business data, commercial areas	medium/context	Optional/manual	Yes if source-backed	Manual/conditional	Community/economic context	Parcels optional; business gap

5. Table Needs

Tables should stay compact in the report body. Detailed feature-level records belong in attachments or the evidence package unless the row count is small and reviewer-approved.

Table 6. Table needs

Table purpose/title	Section	Source data needed	Likely columns	Row type	Body/attachment	Requirement	Current support
Wetlands/waterbodies by comparison unit	3.1.1	NWI, NHD, project/comparison geometries	Unit; stream crossings; wetland class; acres/count; notes; source date	Per comparison unit	Body if compact; attachment if large	Conditional/default if module enabled	Table system exists; needs spec
Wetland/waterbody feature inventory	3.1.1	NWI/NHD intersections	Feature type; name; source ID; unit; acreage/length; direct/buffer flag	Per source feature	Evidence attachment unless small	Optional/supporting	Evidence package likely supports
FEMA flood zones by comparison unit	3.1.2	FEMA/NFHL/DFIRM	Unit; zone; floodway flag; acreage/linear crossing; notes	Per comparison unit + zone	Body if compact	Conditional/default if flood module enabled	Needs policy
Flood hazard feature inventory	3.1.2	FEMA intersections	Zone; BFE if available; floodway; source panel; unit; acreage	Per source feature/zone	Evidence attachment	Optional	Needs table spec
Water quality / watershed context	3.1.3	WBD/HUC, NHD, 303(d), TMDL	HUC; waterbody; direct/downstream; impairment; TMDL status; notes	Summary/waterbody rows	Body if small; attachment otherwise	Conditional	Gap source + table spec
Protected species consultation summary	3.1.4	Agency letters/IPaC/state species	Agency/source; date; finding; conditions; reviewer status	Reviewer-supplied source row	Body or attachment	Manual/conditional	Needs manual material handling
Cultural resource summary	3.2	Authorized MDAH/SHPO/reviewer data	Resource type; count; relation to APE; eligibility/status; caveat	Summary rows; restricted details in attachment	Restricted attachment	Manual/restricted	Needs access policy
Community resources facility table	3.3	Community facility sources	Facility type; name; distance/relation; access sensitivity; source	Per facility	Attachment; compact body list if small	Conditional	Gap
Utility crossings table	3.4.2-3.4.3	Pipeline/transmission/substation sources	Asset type; owner if allowed; crossing/nearby; location unit; coordination flag	Per source feature/crossing	Attachment/body if small	Conditional/manual	Gap
Public water supply wells table	3.4.1	Well/facility source	Facility/well ID; source; relation; protection notes; caveat	Per facility	Body/attachment	Conditional	Gap
Hazardous/regulated facilities risk table	3.5.1	EPA/state databases + reviewer risk rules	Facility; program/source; distance/relation; status; risk tier; reviewer note	Per facility	Body for elevated-risk subset; full list attachment	Conditional/default if hazmat enabled	Source class covered; risk rules gap
Oil/gas wells table	3.5.2	State oil/gas wells	Well ID/name; status; relation; source date	Per source feature	Body if findings; otherwise no table	Conditional	Covered class; state config needed
Income demographics	3.6.1	ACS/Decennial/Census geography	Geography; poverty/low-income metric; comparison geography	Summary rows	Body if compact	Conditional	Gap
Demographic composition	3.6.1	ACS/Decennial	Geography; selected	Summary rows	Body if compact	Conditional	Gap

Table purpose/title	Section	Source data needed	Likely columns	Row type	Body/attachment	Requirement	Current support
			race/ethnicity groups; comparison geography				
Source status / evidence audit	Method/attachment	App source status	Source ID; title; effective truth; acquisition status; sections; caveats	Per source	Evidence package, not report body	Required internally	App likely supports; needs doc shape
Review queue decisions	Evidence/review package	Review queue	Item; evidence; reviewer action; final text state	Per deliverable item	Evidence/review package	Required internally	App supports

6. Alternative / Comparison-Unit Logic

The app should support arbitrary submitted project features/comparison units: routes, corridors, parcels, sites, polygons, study areas, alternatives, or other geometries. Default expansion should be table_only, not narrative children.

Table 7. Comparison-unit behavior

Section	Sample behavior	Generalize?	Recommended Review Assist behavior
Study Area / Project Area	Per-alternative narrative descriptions	Trail/corridor-specific	Use generic comparison-units summary. Support alternatives, parcels, sites, corridors, polygons, etc.
Wetlands/waterbodies	Per-alternative subsection attempted; comparison table used; several subsections blank	Conditional + incomplete	Default to table-only per comparison unit. Narrative children only if reviewer enables and evidence differs materially.
Floodplains/floodways	Comparison table by trail alignment; no per-alt narrative children	Conditional	Table-only per comparison unit by default.
Water quality	Project-wide HUC/downstream context	Generalizable context pattern	Summarize together by watershed; per-unit only when watersheds/receiving waters differ materially.
Protected species	Project-wide agency/range discussion	Manual/conditional	No automatic per-alt expansion unless source findings are unit-specific.
Cultural resources	APE-based, not per alternative	Conditional/manual	APE controls; reviewer defines APEs.
Community resources	Facility/access discussion, not alternative comparison	Conditional	Nearby context list; optionally relate facility to comparison unit.
Utilities/infrastructure	Direct crossings/nearby infrastructure	Conditional	Use table/list by asset; group by comparison unit if direct intersections differ.
Hazmat	Facility/risk list by corridor/buffer	Generalizable hazmat pattern	Use facility-level risk table; optionally relate each facility to comparison unit.
Demographics	Census tracts along corridor	Conditional	Summarize by census geographies; per-unit only if units traverse different communities.
Business nodes	Commercial node narrative	Manual/project-specific	Manual/reviewer-supplied unless POI/business source enabled.

7. Spatial Scope / Extent Semantics

Different extents are allowed and expected. The system must know whether each extent drives direct analysis, nearby/context screening, watershed context, regional context, or presentation-only rendering.

Table 8. Spatial scope and extent semantics

Section	Spatial terms	Implied extent type	Map extent larger?	Table/list alignment	Wording rule
Introduction	project, corridor, planning area	Direct project/study area	Maybe	N/A	Use project area/study area from supplied geometry.
PEL relationship	broader PEL, future NEPA	Manual/reviewer context	N/A	N/A	Keep custom; do not imply every report is PEL-linked.
Study Area	trail alternatives, corridor, terminus, route	Direct comparison-unit extent	Overview likely medium/context	Alternative descriptions not source-table aligned	Use comparison unit generically.
Methodology	within alignments, maps, Appendix panels	Direct + presentation	Yes	N/A	Distinguish analysis extent from map frame.
Wetlands/waterbodies	within, along, adjacent, low-lying	Direct + nearby/adjacent	Likely yes	Table direct-ish but incomplete	Use mapped/intersect/occur within; do not say jurisdictional.
Floodplains/floodways	within, locations, Zone AE, Zone X	Direct project extent	Figure says in/near; analysis says within	Table should use direct intersection only	Separate mapped within project from shown near project.
Water quality	crosses HUC-12s, downstream, subwatersheds	Watershed + downstream	Yes	List is watershed/downstream	Never say impaired waters are within if only downstream.
Protected species	within range, two-mile radius, habitat	Regional/range + buffer + manual	Maybe	Letter findings are not GIS-only	Only state agency/reviewer findings when source supplied.
Cultural	near, APE, within APE, adjacent, visual/noise/vibration	Manual APE + nearby context	Cultural maps may be restricted	Must match APE definition	Public context is not authorized SHPO/MDAH records.
Community resources	along, near, just outside, within or near	Nearby/community context	Yes	Lists mix direct and nearby	Label within vs near/outside; avoid exhaustive claims.
Utilities/energy	intersect, cross, nearby	Direct + nearby	Map may be context	List direct and nearby	Label crossing vs nearby asset; require owner coordination.
Hazmat	within, 0.1 mile, adjoining, nearby	Screening buffer + adjacency	Yes	Counts use multiple extents	Every count must carry its extent.
Oil wells	within project limits	Direct extent	No unless figure	No-finding direct only	Do not imply broader county absence.
Demographics	census tracts along, citywide, county	Census/community/regional	Yes	Tables use tracts + city/county comparisons	Census context, not direct impact area.
Business nodes	along corridor, commercial district	Manual/community context	Context	Narrative only	Manual/reviewer context unless source-backed.

Extent language rules

Table 9. Extent wording rules

Term	Use only when
within	Feature geometry intersects the project/comparison-unit extent or a named buffer stated in the sentence
near	Feature is outside direct extent but within a configured screening distance
adjacent	Parcel/feature touches or is explicitly tagged adjoining; do not use for generic nearby
in the vicinity	Avoid unless extent is named
downstream	Hydrologic network or reviewer confirms flow relationship
watershed/subwatershed	HUC/WBD context drives the analysis, not direct footprint
APE	Reviewer defines cultural APE; do not infer from generic buffer unless policy says so
county/city/regional	Demographics/species/airport context; never imply direct project impact
corridor	Only if project geometry is a corridor/route; otherwise use project area or comparison unit
shown on map	Figure support only; not evidence of direct intersection

8. Wording and Caveat Library

Wetlands / NWI

Reusable pattern: Desktop screening identified mapped wetlands and/or waterbodies from configured sources within or near the named extent. These mapped features are useful for early planning but do not establish jurisdictional limits. Final boundaries and jurisdictional status require field delineation and agency coordination.

Avoid: jurisdictional wetlands are present; no wetlands are present; NWI confirms boundaries; impacts will or will not occur.

Hydrography / water quality

Reusable pattern: Hydrography and watershed datasets identify streams, drainage crossings, receiving waters, and watershed context. If impaired waters are downstream rather than intersecting the project, describe them as downstream/contextual.

Avoid: impaired streams are within the project when only downstream; project will affect water quality without design facts; no water-quality concerns based only on no direct 303(d) intersections.

Floodplain / floodway

Reusable pattern: FEMA flood hazard data identify mapped flood zones/floodways within the named extent. Work in mapped flood hazard areas may require local floodplain review and hydrologic/hydraulic analysis depending on final design.

Avoid: permit required unless confirmed; no flood risk; minimal hazard means no drainage concern.

Protected species / critical habitat

Reusable pattern: Protected species and habitat findings are based on named source/agency material. Desktop range or database screening does not by itself determine project effect.

Avoid: no effect; no species present; no suitable habitat unless agency/reviewer-confirmed.

Cultural resources / restricted records

Reusable pattern: Separate authorized agency/reviewer-supplied records from public/coarse screening context. Sensitive details should remain in restricted evidence unless approved for disclosure.

Avoid: public/coarse data as SHPO/MDAH records; no historic properties affected without Section 106 basis; all archaeological resources identified.

Community resources

Reusable pattern: Identified nearby community resources may be sensitive to access, detours, noise, or construction staging. Lists are planning-level screens and should be verified with local stakeholders where operations may be affected.

Avoid: all facilities; safe access will be maintained unless committed.

Regulated / hazardous facilities

Reusable pattern: Regulatory database screening identifies facilities within the named extent/buffer. Database presence indicates possible due-diligence considerations and does not establish contamination, liability, cleanup status, or construction risk.

Avoid: site is contaminated without source support; no contamination risk; Phase I ESA conclusion without a Phase I or hazmat report.

Utility / infrastructure

Reusable pattern: Mapped utility and infrastructure features are planning coordination inputs. Locations and ownership should be verified with providers before design or construction; sensitive details require access controls.

Avoid: conflict resolved; utility location confirmed; service disruption likely without design evidence.

Socioeconomic / demographics

Reusable pattern: Census data provide community context for planning and outreach. Census geographies may extend beyond the project area, so tract-level data should not be described as direct project-area demographics without qualification.

Avoid: environmental justice impact; disproportionate effect; all affected residents.

Screening-only / no determination

Reusable pattern: The report is a screening-level constraints review. It does not replace field surveys, agency determinations, permitting, design-level impact analysis, legal conclusions, or regulatory findings.

Avoid: compliance determination; final impact conclusion; binding commitment.

9. Manual / Reviewer-Supplied Materials

Table 10. Manual and reviewer-supplied materials

Component	Why manual/reviewer-supplied	Recommended app handling
Relationship with PEL Study	Parent-study relationship is project context	Manual section; conditional
Project purpose/background	Usually project narrative, not GIS-derived	Reviewer-supplied metadata/narrative
Alternatives narrative	Design/planning context	Comparison-unit metadata; no GPT facts unless supplied
Agency consultation letters	External agency records	Upload as manual source; summarize only after review
Protected species findings	Agency/reviewer confirmation needed	Manual/attachment-backed
SHPO/MDAH cultural records	Restricted/authorized records	Restricted source type; no public substitution
APE definition	Reviewer/cultural specialist defines	Required manual geometry/notes
Hazardous Materials Assessment Report	Specialist report/risk ranking	Attachment + reviewed summary
Hazmat risk categories	Requires policy/specialist judgment	Review-gated; no GPT-only ranking
Utility owner coordination	Provider/design coordination	Manual flags + owner contact notes
Business/economic node narrative	Local context often not dataset-complete	Manual/reviewer-supplied
Prior studies	Project record	Manual source upload
Field verification status	Human fieldwork	Manual evidence
Final conclusions/next steps	Reviewer judgment	Review-gated; no new facts

Component	Why manual/reviewer-supplied	Recommended app handling
Commitments/mitigation	Agency/project commitments	Manual only

10. Generalizable vs Project-Specific

Table 11. Generalizable vs project-specific extraction

Bucket	Elements	Action
Generalizable template elements	Cover metadata, executive-summary shape, TOC/list generation, intro, methodology, source/data procedures, limitations, resource-category inventory, references, attachment list, map/table/evidence model, caveat library	Keep as generic structure
Conditional elements	Wetlands, floodplains, water quality, protected species, cultural, community resources, utilities, contamination, socioeconomic, business nodes, panel maps, hazmat attachment, agency letters	Include by source/project triggers
Project-specific elements	Project names, place names, named facilities, exact counts, addresses, dates, local creek/facility descriptions	Do not copy into reusable policy
Trail/corridor-specific elements	Trail alternatives, stream crossings by alignment, rail/highway/nature corridor language, panel maps split by corridor sections, boardwalk/culvert trail wording	Support only as comparison-unit/corridor mode
PEL-specific elements	Relationship with PEL Study, PEL team framing, transition into NEPA from PEL, PEL public/stakeholder process language	Conditional/manual; not default
Incomplete/uncertain elements	Blank table cells, unresolved comments, broken references/figure numbering, placeholder attachments, mixed project identities	Mark draft-only; do not generalize
Do-not-generalize elements	Exact regulatory conclusions, agency findings, exact facility lists, private-facility PEL rule, ArcGIS version, no-significant-impact language	Ban or review-gate

Specific call: Relationship with the PEL Study

Do not include this section in the default Review Assist template. Treat it as `intro.parent_study_relationship` with `include_default false`, `drafting_mode manual_reviewer_supplied_only`, and a trigger such as `parent_study.type` in [PEL, planning_study, corridor_study].

11. Diff Against Current Review Assist App

Table 12. Diff against current app context

Target item	App diff classification	Notes
Generic report structure	App has export package but needs policy/matrix	Add canonical section policy
Source status / effective truth	App covers this well	Surface in methodology/references
Compact report body + evidence package	App covers this well	Preserve; no raw dumps
Wetlands/NWI	App has source	Needs section policy, caveat, table spec
Hydrography/NHD	App has source	Needs crossing metrics and waterbody table policy
FEMA flood hazard	App has source	Needs floodway/flood-zone metrics, table, caveats
Water quality 303(d)/TMDL	App has hydro source but lacks water-quality source	Add state impaired-waters/TMDL source class
HUC/WBD watershed context	Partial/unclear	Add explicit watershed source and policy
Protected species/critical habitat	App lacks needed source	Add future/manual/IPaC/critical habitat/state species support
Agency consultation letters	Manual/reviewer-supplied only	Add manual source type + attachment summary workflow
Public/coarse cultural context	App has source class	Needs strict wording guard

Target item	App diff classification	Notes
Restricted MDAH/SHPO	Future placeholder	Needs access controls + manual source policy
APE logic	App likely lacks policy	Add reviewer-defined APE geometry/material
Community resources	App lacks source/section	Add optional modules or defer
Public water supply wells	App lacks source	Add later state/EPA source
Utility/pipeline/electric infrastructure	App lacks source	Add manual/sensitive/deferred first
Airports	App lacks source	Add FAA source if useful
Hazmat database screening	App covers source class	Needs risk/ranking and attachment policy
Oil/gas wells	App has source class	Needs state config and no-finding wording
Demographics/Census/ACS	App lacks source	Add TIGER/ACS/Decennial module
Business/economic nodes	Manual/reviewer-supplied only	Optional parcels/POI later
Figures	App has figure generation	Needs figure matrix, extent classes, panel policy
Tables	App has deliverable tables	Needs per-section table specs
Comparison-unit expansion	App needs policy	Default table only; narrative only if triggered
GPT Interpretive Assist	App wiring is right	Needs per-section drafting modes/prohibited claims
DOCX export quality	App exports but needs QA gate	Add caption/cross-ref/empty-table/comment/render checks

12. Recommended Implementation Moves

Table 13. Implementation moves

Priority	Move	Why
P0	Create canonical review_assist_report_policy.json	Provides section/source/figure/table/caveat policy spine
P0	Make extent_type and visual_extent_class first-class	Prevents map extent from becoming analysis claim
P0	Add comparison-unit expansion guard	Default table_only; narrative children only when enabled
P0	Add caveat and prohibited-claim libraries	Keeps deterministic and GPT drafting aligned
P0	Add source-needs manifest by section	Prevents sections from rendering without adequate sources
P0	Add source truth/status wording	Labels incomplete, stale, deferred, manual data honestly
P0	Segregate cultural source classes	Public/coarse context must not become authorized SHPO/MDAH records
P0	Add manual/restricted source types	Needed for letters, hazmat reports, cultural records
P0	Add GPT eligibility matrix	GPT only where evidence + policy + caveats exist
P0	Add export QA checks	Catch empty tables, comments, broken captions, figure numbering, stale refs
P1	Build figure/table matrix from policy	Makes deliverables predictable and testable
P1	Add wetland, flood, waterbody table specs	High-value and already source-backed
P1	Add hazmat facility table/risk review workflow	Current sources exist but risk wording needs guardrails
P1	Add water-quality context policy	Avoids wrong direct/downstream language before full source expansion
P1	Replace sample Study Area with neutral Project Area and Comparison Units	Keeps app feature-neutral
P1	Add attachment manifest	Tracks maps, evidence package, manual attachments, agency letters
P1	Connect review queue to sections/tables/figures	Human decisions stay explicit
P2	Add WBD/HUC + state 303(d)/TMDL sources	Needed for water-quality module
P2	Add species/critical habitat/manual agency letter support	Needed for protected species
P2	Add Census/ACS module	Needed for socioeconomic section
P2	Add community facilities and public water supply sources	Needed for community/infrastructure sections
P2	Add utility/energy/FAA sources as manual/sensitive/deferred	Useful but needs careful source policy
P3	Add GPT style context file	Tone/shape only; no project facts
P3	Polish report rendering, captions, TOC, panels, glossary	Better UX after core policy works

GPT pilot recommendations

Table 14. Lower-risk GPT pilot sections

Good GPT pilot	Why
Methodology source summary	Deterministic source list can bound it
Limitations/Data gaps	Caveat library controlled
Wetlands/NWI summary	Strong source + strong caveat
FEMA floodplain summary	Strong source + deterministic table
Hazmat database context	Useful if risk labels are reviewer-gated
Community resource context	Useful but must say identified nearby
Demographic context	Good if no EJ/disproportionate-impact conclusions

Avoid early GPT for cultural determinations, protected species effect language, agency consultation interpretation, hazmat risk rankings, final commitments, PEL relationship, and business/economic impacts.

13. Suggested Machine-Readable Policy Shape

Use one canonical file: config/review_assist_report_policy.json, plus one concise explanatory doc: docs/domains/REPORT_POLICY.md. Avoid premature config sprawl.

- Sections carry trigger, extent, visual extent, comparison-unit behavior, evidence patterns, drafting mode, caveats, prohibited claims, allowed refs, and review gates.
- Sources carry source class, public/restricted/manual status, effective truth, stale/deferred status, and section support.
- Figures and tables are generated from policy, not ad hoc report text.
- GPT eligibility is section-specific and bounded by evidence and prohibited-claim bundles.

14. GPT Style Context Recommendations

The GPT style context should guide tone and shape only. It should not contain project facts, place names, counts, facility names, agency findings, dates, trail/PEL-specific content, or unfinished draft content.

Include

- Compact environmental constraints prose
- Source-backed, cautious language
- Explicit extent semantics
- Caveats for public, coarse, incomplete, stale, manual, or restricted data
- References only to allowed tables, figures, sources, and attachments

Exclude

- Project facts from example reports
- Trail-specific or PEL-specific language unless active project context supports it
- Place names, counts, facility names, addresses, agency findings, and dates from example reports
- Final impact, effect, eligibility, contamination, permit, or commitment claims

GPT style mini-template

Write concise, screening-level environmental constraints prose.

Use only the evidence supplied in the section evidence packet. Do not infer missing source findings, regulatory determinations, design commitments, permit requirements, cultural eligibility, species effect, contamination status, or final impacts.

Preserve extent semantics:

- within = direct intersection with named analysis extent
- near = within configured screening buffer
- downstream = hydrologic relationship supported by source or reviewer
- watershed/subwatershed = contextual hydrologic geography
- APE = reviewer-defined cultural extent
- shown on figure = presentation support, not impact evidence

Use cautious phrases such as desktop screening identified, mapped features indicate, based on available source data, may require field verification or agency coordination, and should be reviewed as design advances.

Avoid confirmed, no impact, no effect, jurisdictional, eligible/ineligible, all facilities, safe access ensured, permit required, contamination present/absent, and disproportionate impact.

Keep the report body compact. Refer detailed records to tables, figures, attachments, or the evidence package.

15. Codex Prompt Starters

P0 policy object prompt

Build config/review_assist_report_policy.json as the canonical report policy object. Every section must define include_default, trigger, extent_type, visual_extent_class, comparison_unit_expansion, evidence_patterns, drafting_mode, required_caveat_bundles, prohibited_claim_bundles, allowed_sources, allowed_tables, allowed_figures, and review requirements. Update report generation so a section cannot render unless its trigger, source status, extent type, caveat bundle, and allowed refs are satisfied.

PO extent semantics prompt

Add explicit analysis_extent and visual_extent fields to deliverable sections, figures, and evidence records. Update wording helpers so within, near, adjacent, downstream, watershed, county/regional, APE, corridor, and shown on map are only used when their configured semantics are satisfied. Add tests preventing a watershed or visual map extent from being described as a direct project intersection.

PO GPT guardrail prompt

Implement a GPT eligibility matrix from report policy. GPT drafting is allowed only for sections whose policy permits it and whose evidence packet contains the required source-backed records. Enforce prohibited claim bundles before and after generation. Block or review-gate final impact, no effect, jurisdictional, eligibility, contamination, permit, and commitment language.